NETVISOR Mini Discovery V3

Project Information

Project Name: NETVISOR

Module: Discovery Engine

Version: V3

Author: Moad

Date: 2025

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1. Overview

NETVISOR Mini Discovery V3 extends the discovery capabilities introduced in V1 and V2 by adding TCP-based service detection.

While previous versions focused on identifying active hosts and collecting MAC addresses, V3 introduces the ability to detect open network services through TCP port probing.

This enhancement allows the system to begin understanding the role of discovered devices within the network.

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2. Objectives

The objectives of V3 are:

- Maintain active host discovery
- Maintain IP-to-MAC correlation
- Detect open TCP ports
- Identify accessible services
- Enrich discovered assets
- Prepare for service inventory and asset classification

3. Architecture

Directory Structure

netvisor-mini-discovery-v3/

core/

- Network.php
- Scanner.php

modules/

- Ping.php
- Arp.php
- Tcp.php

services/

- DeviceBuilder.php

models/

- Device.php

run.php

4. Components

4.1 Ping Module

Responsibilities:

- Discover active hosts
- Validate network reachability

Technology:

- ICMP Echo Requests

4.2 ARP Module

Responsibilities:

- Collect Layer 2 information
- Retrieve MAC addresses

Technology:

- Linux Neighbor Table
- ip neigh

4.3 TCP Module

New component introduced in V3.

Responsibilities:

- Probe selected TCP ports
- Detect open services
- Enrich device information

Default ports scanned:

- 22 (SSH)
- 80 (HTTP)
- 443 (HTTPS)

Functions:

- scanPorts()

Technology:

- fsockopen()

4.4 Device Model

Stores:

- IP Address
- MAC Address
- Status
- Open Ports

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5. Discovery Workflow

Phase 1

Network Detection

Example:

192.168.1.0/24

Phase 2

Ping Sweep

Discover active hosts.

Output:

Active IP addresses

Phase 3

ARP Collection

Retrieve:

IP → MAC

Mappings.

Phase 4

TCP Discovery

For each active host:

- Test port 22
- Test port 80
- Test port 443

Record open ports.

Phase 5

Asset Enrichment

Combine:

- IP Address
- MAC Address
- Open Ports

Into a structured device record.

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6. Features Implemented

Implemented in V3:

✓ Single Network Discovery

✓ ICMP Ping Sweep

✓ ARP Analysis

✓ MAC Address Collection

✓ TCP Port Discovery

✓ Service Detection

✓ Device Enrichment

✓ Modular Architecture

✓ CLI Execution

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7. Example Output

Example:

IP: 192.168.1.105

MAC: 0e:24:2c:a3:3e:23

PORTS: 80,443

IP: 192.168.1.254

MAC: e0:3d:a6:95:4b:50

PORTS: 80

Total Devices: 8

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8. Improvements Over V2

Feature	V2	V3
Active Host Discovery	✓	✓
MAC Address Collection	✓	✓
ARP Analysis	✓	✓
TCP Discovery	✗	✓
Open Port Detection	✗	✓
Service Visibility	✗	✓
Asset Enrichment	Limited	Enhanced

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9. Data Collected

V3 device inventory now contains:

- IP Address
- MAC Address
- Status
- Open Ports

Example:

Device

IP: 192.168.1.105

MAC: 0e:24:2c:a3:3e:23

Ports:

- 80
- 443

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10. Current Limitations

Not implemented yet:

✗ Hostname Resolution

✗ DNS Reverse Lookup

✗ Vendor Identification (OUI)

✗ Operating System Fingerprinting

✗ Service Banner Grabbing

✗ Database Storage

✗ Multi-Network Discovery

✗ Topology Mapping

✗ Monitoring Engine

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11. Challenges Encountered

During development, several technical challenges were identified:

- Timeout handling during TCP probing
- Devices blocking ICMP responses
- Missing MAC addresses in ARP cache
- Service detection accuracy limitations
- Scan speed versus network size trade-offs

These challenges helped establish a foundation for future optimization.

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NETVISOR MINI DISCOVERY V3
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Phase 1: Ping Sweep...
Phase 2: ARP Table...
===== DEVICES FOUND =====

IP: 192.168.1.2
MAC: unknown
PORTS: 80, 443

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IP: 192.168.1.15
MAC: unknown
PORTS: none

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IP: 192.168.1.43
MAC: unknown
PORTS: none

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IP: 192.168.1.56
MAC: unknown
PORTS: none

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IP: 192.168.1.68
MAC: unknown
PORTS: none

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IP: 192.168.1.76
MAC: unknown
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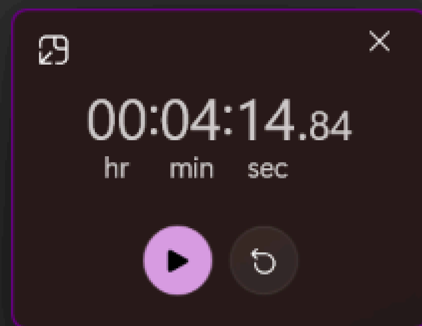

12. Future Roadmap

Version 4

Hostname & Vendor Detection

- Reverse DNS Lookup
- Hostname Collection
- Vendor Identification via MAC OUI

Version 5

Multi-Network Discovery

- Multiple Subnet Support
- Network Registry

Version 6

Database Layer

- Asset Storage
- Historical Tracking

Version 7

Topology Engine

- Router Detection
- Network Mapping
- Relationship Discovery

Version 8

Monitoring Engine

- Health Checks
- Metrics Collection
- Alerting

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13. Conclusion

NETVISOR Mini Discovery V3 successfully transforms the project from a simple host discovery tool into an asset enrichment platform.

By combining ICMP discovery, ARP analysis, and TCP service detection, the system now provides a more complete view of network devices and their exposed services.

This version establishes the foundation for future asset inventory, topology mapping, and monitoring capabilities within the NETVISOR platform.